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ORIGINAL PAPER

Early-stage of invasion by beech bark disease does not necessarily trigger American beech root sucker establishment in hardwood stands

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Abstract Two concomitant phenomena currently affect the dynamics of sugar maple-American beech (AB) stands in northeastern North America: beech bark disease (BBD), and increased AB understory density. Many studies suggest a causal link between the two phenomena, i.e., BBD favouring beech regeneration. But this link has yet to be experimentally demonstrated. To address the question, we compared regeneration composition between recently BBD-affected and -unaffected stands. A total of 109 stands were sampled; half were affected by BBD. Seedling and sapling density were assessed, together with the origin (seedling or sprout). While BBD affects stands in the eastern part of the study region, AB was observed in the understory across the entire region. No clear difference in AB sprout density between BBD-affected and -unaffected stands was observed while AB seedling density—as well as pooled AB seedling and sprout density were higher in unaffected stands. Findings suggests that BBD, in its early stage, is not a necessary trigger of AB understory establishment. Yet, AB sapling basal area generally was higher in stands affected by BBD, likely indicating a greater rate of AB understory development due to increased light availability beneath a more open crown canopy. That development can lead to AB understory dominance. This distinction—BBD not necessarily triggering AB root sucker establishment but favoring AB advance regeneration development—also questions the generalized perception that dense AB thickets necessarily originate from root suckers.

Keywords Beech bark disease · American beech regeneration · Sugar maple regeneration · Beech sprout

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Introduction

Regeneration dynamics of sugar maple (SM, *Acer saccharum* Marsh.) and American beech (AB, *Fagus grandifolia* Ehrh.) have received a great deal of attention over the past several decades across a wide range of the respective species distributions (e.g., Brisson et al. 1994; Canham et al. 1994; Runkle 2013; Woods Woods 1979). While these species appear to co-exist (Poulson and Platt 1996), other authors have reported either a strong decadal increase in AB regeneration (Duchesne et al. 2005; Hane 2005; Gravel et al. 2011) or a SM regeneration failure associated with an over-abundance of AB regeneration (Nolet et al. 2015). Hereafter, we refer to this phenomenon as the "AB understory dominance issue" that is characterized by thousands of AB saplings and poles per hectare (Duchesne et al. 2005; Nolet et al. 2008).

Bose et al. (2017) has identified many factors that may contribute to AB understory expansion in SM-AB-dominated stands. We define AB expansion as an increase in regeneration establishment (seedlings or sprouts) or development (growth and survival) of advance regeneration. Among these factors, a higher shade tolerance of AB compared to SM is associated with a greater potential for the former to survive and develop in a shaded understory (Canham 1990), and for increased growth and development in the brightened environment after a canopy disturbance (Beaudet et al. 2002; Nolet et al. 2008). Other studies have pointed towards greater AB tolerance to poor soils (Kobe et al. 2002), or even suggest an allelopathic effect of AB over SM (Hane et al. 2003). The capacity of AB to produce sprouts (Forcier 1975) allows this species to colonize understories with low light-levels. These grow faster growth than seedlings (Beaudet and Messier 2008). While all of these factors can contribute to AB understory expansion, none alone can explain why a drastic change would occur in AB understory dynamics.

At the same time, another important phenomenon that can affect their regeneration dynamics is occurring in SM-AB stands: the invasion of beech bark disease (BBD) (Houston 1994; Cale et al. 2017). BBD is a disease complex formed by the interaction between an insect, (beech scale, *Cryptococcus fagisuga* Lindinger) introduced around 1890 in Halifax (Hewitt 1914), and two native pathogens (*Neonectria ditissitima* and/or *N. faginata)*. American beech scale (*Xylococculus betulae* [Pergande]) and *N. ditissima* (Tul. & C. Tul.) also native species, contribute to this complex disease process. Recent work (Cale et al. 2017) also shows that birch margarodid (Xylococculus betulae), a native species, may also contribute to the BBD complex. The sap-feeding activity of beech scale facilitates further infection of phloem and cambial tissue by the pathogen (Shigo 1972). The infection affects the infected trees and can eventually kill them.

The development of BBD within stands progresses through three phases (Shigo 1972; Houston 1975; Cale et al. 2017): (1) the advancing front, where beech scale has affected the trees, but infection by the pathogen is rare; (2) the killing front, where high scale-insect infestations result in high levels of *Neonectria* infection and mortality within the stand; and (3) the aftermath stage, where stands retain low densities of both the pest and pathogen. Through time, BBD spread from the point of origin in Nova Scotia (Halifax), westward and southward into Ontario and the USA during the 20th C. (Cale et al. 2017), and is now spreading into southwestern Québec (Nolet et al. 2015).

Various studies have observed an increase in AB understory (saplings and poles) after BBD infestation (e.g., Twery and Patterson 1984; Garnas et al. 2011). Of these, Garnas et al. (2011) observed an increase of about 350% in AB understory density after infestation by BBD. BBD also has been suggested to increase sprout production (Houston 1975). Taken altogether, this information may lead to the perception that BBD triggers the development of dense beech thickets through a root-sprouting response (e.g. Cale et al. 2013). Also, it is not clear whether such response would occur while the trees are losing their vigor due to a BBD infection, or once they are dead. Hence, despite the many studies on AB regeneration and BBD, it still remains unclear whether BBD actually triggers beech understory dominance and, therefore, how these two phenomena might interact to affect SM-AB regeneration dynamics. In fact, available information indicates that other causes trigger root suckering in AB (e.g., damage to shallow roots during logging, and from other surface disturbances). Also, several publications reported an abundance of understory beech in stands never affected by logging (e.g., Ward 1961; Held 1983; Hough 1937), even in uncut old-growth stands (Beaudet et al. 1999). In addition, the publication by Curtis and Rushmore (1958) described development of a dense and interfering AB understory in harvested stands prior to the time that BBD became recognized as a factor within New York's Adirondacks. Richards and Farnsworth (1971) also reported negative effects of pre-existing understory beech on the regeneration of other species after a range of overstory cutting treatments. So even with forests not yet infected by BBD, these and other sources suggested that managers needed to account for interference by AB root sucker understories as part of regeneration programs in upland hardwood stands.

Quebec's Outaouais region offered the opportunity to assess whether AB understory development and BBD are intrinsically linked, given that BBD is currently spreading in the region and large forest tracks remain unaffected by the disease. The main objectives were (1) to test whether AB understory expansion is triggered or favoured by BBD in its early stage, (2) to investigate possible differences in AB regeneration origin (sexual vs. vegetative) between areas affected by and free from BBD.

Methods

Study area

The study was located in the Outaouais region (30,472 km^2) of southwestern Quebec (Canada), within the western sugar maple–yellow birch (*Betula alleghaniensis* Britton) bioclimatic region (Saucier et al. 2009). The study sites cover 45°42′58″–46°46′53N by 77°45′42″–74°49′3″W. This landscape has numerous hills, with respective maximum and mean elevations of 450 and 300 m above sea level (Robitaille and Saucier 1998). Mean annual temperature is 3.7 °C, with an annual precipitation of about 1000 mm (including 250 mm falling as snow). There are 2716 degree-days above 0 °C (Environment Canada 2016). Soils of the study area are characterized by thin to moderately thin glacial till parent material, composed of metamorphic rocks, such as gneiss. Stand composition is typically a mix of SM, AB, yellow birch, American basswood (*Tilia americana* L.), ironwood or American hop-hornbeam (*Ostrya virginiana* [Mill.] K. Koch), eastern hemlock (*Tsuga canadensis* [L.] Carrière) and balsam fir (*Abies balsamea* [L.] Miller), and depends upon slope and drainage.

Stand selection

Based on Quebec government temporary sampling plots located in the Outaouais region, we pre-selected plots according to the following criteria: (1) having at least 50% of the basal area in sugar maple and American beech; (2) having at least one tree > 9.1 cm DBH of each of the two species. More than 3000 plots met these criteria. Among these, a stratified random sampling method was subsequently used to select at least 1 plot (stand) in each of the 64 different sections of the region, based upon latitude and longitude. Moreover as an additional criterion in stand selection, we ensured, that our sampling covered a variety of beech understory conditions (as represented by saplings > 2 cm and < 9.1 cm in DBH) ranging from low (0–500 stems ha^{-1}) to medium (501–3500 stems ha^{-1}) and high (> 3500 stems ha^{-1}) beech sapling densities. A total of 109 stands, based on the Quebec government plots, were selected for further sampling (Fig. 1).

Data collection

Data were collected from spring to autumn 2014. In each stand, we positioned one centre point, plus four complementary points located 7 m from the centre in the four cardinal directions. The centre point was used to establish two distinct plots. The first was a basal area prism (metric BAF 2) sweep to estimate SM, AB and total stand basal area (m^2 ha^{-1}) for trees ≥ 9.1 cm in DBH. The second was a circular plot (400 m^2) in which BBD infection level was recorded for each beech tree, as described in Table 1. The four complementary points were used as sub-plots for sampling seedlings (height > 10 cm and DBH < 2 cm) and saplings (2 cm ≤ DBH, < 9.1 cm), which were respectively of 5 and 20 m^2 in area. In each sub-plot, the number of individuals per species was recorded. Two AB seedlings also were dug out in each subplot to identify their origin (sexual or vegetative propagation). The ratio seedlings/sprouts obtained from the 4 sub-plots (8 stems per plot) was then attributed to the total number of stems (seedlings and sprouts) in the plot. Overall more than 400 seedlings were dug out in the study for both stands with and without BBD, representing about the half of AB stems we tallied. To avoid confusion in terms for AB hereafter, we used the terms "seedling" to designate a stem of sexual origin and "sprout" as a stem of vegetative origin that meet the size criteria of a seedling. It is worth mentioning that such a distinction (sexual vs. vegetative origin) was not made for saplings. Light percentage was estimated with a densiometer at 1.3 m height at each corresponding plot centre. Two soil samples were taken from the B-horizon and pooled to perform pH analyses. Soil pH was measured in deionized water using a 1:2 ratio and was measured using a pH/ion

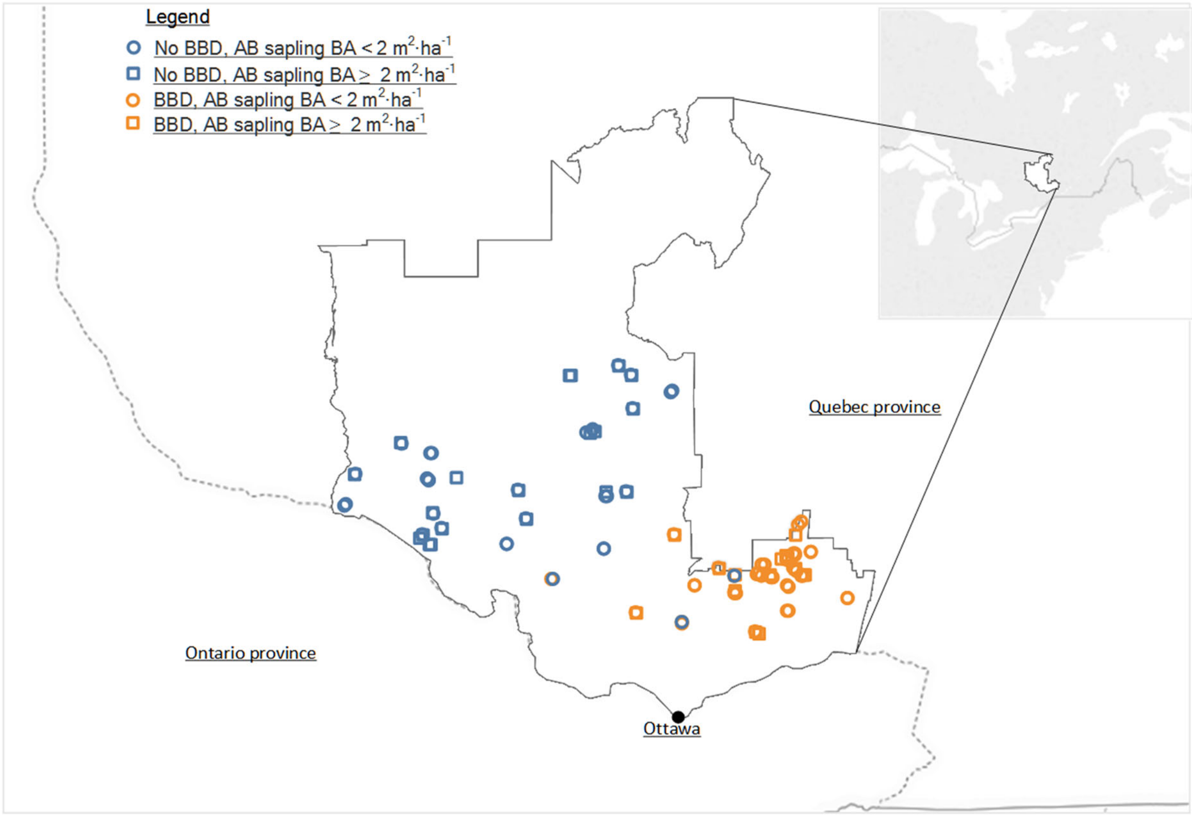

Fig. 1 Distribution of stands that were sampled within the study region, together with their BBD and understory beech characteristics

Table 1 Description of BBD infection level at the stem scale

Infection level	Description
1	Tree without any sign of BBD
2	Tree infected by beech scale only; generally > 75% of the crown remaining
3	Tree infected by beech scale and and Neo*nectria*; between 25 and 75% of the crown remaining
4	Tree infected by beech scale and and Neo*nectria*; < 25% of the crown remaining
5	Dead, with signs of BBD
6	Dead with no sign of BBD

meter S220 (Mettler-Toledo AG, Schwerzenbach, Switzerland) (Hendershot et al. 2007).

Data analyses

Statistical analyses followed the model comparison approach based upon the Kullback–Leibler information quantity as presented by Anderson et al. (2000). For each response variable, we first compared the performance (using the Akaike information criterion, AIC_C) of all logical model combinations, which ranged from single effects to double and triple interactions among our predictor variables (See Table S1, supplementary material). If this first step identified models with different predictor variables but similar AIC_C scores, a second round of model comparisons was performed using the variables that were included in these prior models. The response variables were: AB seedling density, AB sprout density, pooled AB seedling and sprout density, AB sapling basal area. The basic predictor variables that were tested included pH, slope, harvesting history (no harvesting, old partial cuts—before 1995—or recent partial cuts—1995 to 2010, as provided by provincial forest maps), AB basal area of trees DBH > 9.1 cm and BBD infestation. Our analyses were performed using BBD infestation in terms of presence/absence, since our other attempts to describe (e.g. Shigos' stage) or quantify BBD (e.g. AB mortality percentage) at the plot level did not lead to higher predictive power than presence/absence. BBD was consider present as soon a tree in the stand showed signs of infection by beech scale.

For each response variable, the difference in AICc between the most parsimonious model and each of the other models was used to calculate the model's weight (ω), which can be interpreted as the probability that a particular model is the best among all candidate models (Anderson et al. 2000). Since a predictor variable may appear in several tested models, it is also possible to sum the weights of the models in which a predictor variable appears. This cumulative weight (ω_c) can be interpreted as the probability that a specific predictor variable is part of the best-tested model. Overall, this analytical approach avoids discarding a predictor variable only because it is not part of the best model. Moreover, as the information-theoretic (based on AIC) and frequentist (based on *P* values) approaches represent two different paradigms, Anderson et al. (2001) clearly recommend not reporting *P* values when using the information-theoretic approach. However, all our models were compared to the intercept-alone model to make sure that the selected model(s) explains more variability than the intercept alone. Model assumptions were evaluated and data were log-transformed when required. All analyses were performed using the *lm or glm* function in R (version 3.2.3; R Core Team 2015).

Results

Almost half (48%) of the stands that we sampled were still not affected by BBD in 2014 (Table 2), as BDD had not spread across the entire Outaouais region. Rather, it was frequent in the southeastern part, and absent in the western part (Fig. 1). When present, BBD was mostly between the advancing and killing phases (sensu Shigo 1972), given that clear signs of vigor loss in crowns were observed (stages 3–5, Table 3) while mortality was still limited. This is because of this low mortality rate that we consider that our region is still in the early-stage of BBD. Beech understory dominance (See Fig. S1, supplementary material) was observed across the entire Outaouais region (Fig. 1). About 40% of the plots had a high beech understory cover in both BBD affected and unaffected stands (> 2 m^2 ha^{-1} for saplings, Fig. 1).

Pooled beech seedling and sprout density was clearly positively affected by TB_{AB} and negatively affected by BBD as shown by models 1 ($\omega = 87\%$) and 2 ($\omega = 12\%$) in Table 4. When beech parent trees (≥ 9.1 cm in DBH) were taken into account, regeneration was 1.32-fold more abundant in unaffected stands than in stands affected by BBD (Fig. 2). This difference in is mainly attributed to seedlings. Model 2 also demonstrated that beech seedling density may also be positively related to pH.

The models that best explained beech regeneration density through vegetative propagation were different from those for sexual origin (Table 4). TBA_{AB} alone provided the best model ($\omega = 69.8\%$, model 8, Table 4) for explaining AB sprout density. The weight (ω) of other models that were tested was $\leq 10\%$, except for the model that included slope with TBA_{AB} ($\omega = 15.9\%$, model 10). Interestingly, BBD was not a predictor in these models (Table 4). Moreover, sprout density was not related to BBD-related mortality (Fig. 3).

Model comparisons showed that the best model explaining AB seedling density included BBD, TBA_{AB}, light avaialbility, slope and pH ($\omega = 73\%$, model 7, Table 4). BBD was the variable with the highest cumulative individual weight ($\omega = 99\%$), while the four other variables had cumulative individual weights ranging from 73 to 95%. AB seedling density increased with light availability, TBA_{AB} and pH, but decreased almost by half when BBD was present in the stand (Fig. 2).

The best model that explained variation in beech sapling basal area (SBA_{AB}) was the interaction between BBD, slope and harvesting history ($\omega = 100\%$, model 13, Table 4). SBA_{AB} was lower in stands where no harvesting had been performed, while it was greatest in old partial cuts (Fig. 4). A positive relationship between BBD and SBA_{AB} was

Table 2 General description of BBD-affected and -unaffected sampled stands

Descriptor	Unaffected stands	Affected stands
Number of stands	52	57
Mean total BA (m^2 ha^{-1})	24.9 (5.3)	23.4 (5.2)
Mean SM BA (m^2 ha^{-1})	9.7 (6.5)	11.5 (6.4)
Mean AB BA (m^2 ha^{-1})	9.5 (6.1)	6.1 (4.9)**
Mean SM pole (m^2 ha^{-1})	2.2 (3.2)	2.5 (2.9)
Mean pole-AB (m^2 ha^{-1})	2.9 (3.1)	3.0 (2.9)
Mean DBH (cm)	20.9 (4.6)	20.5 (4.8)
Mean AB mortality (%)	0.04 (0.16)	0.16 (0.20)***
Mean elevation (m)	324.5 (37.9)	329.1 (50.2)
Mean slope (%)	4.5 (2.3)	6.7 (3.4)***
Mean pH	5.0 (0.3)	4.8 (0.4)

Numbers represent average by stands and their corresponding standard deviations (in parentheses)

** $p < 0.01$; *** $p < 0.001$

Table 3 BBD tree infection level proportions (in basal area) by DBH classes for stands with and without BBD

BBD presence	DBH class (cm)	Tree infection level					
		1	2	3	4	5	6
No	9.1–19	0.99					0.01
	19.1–29	0.96					0.04
	29.1–40	0.94					0.06
	> 40	0.90					0.10
	Total	0.95					0.05
Yes	9.1–19	0.55	0.40	0.01	0.01	0.00	0.01
	19.1–29	0.24	0.48	0.12	0.06	0.05	0.04
	29.1–40	0.12	0.38	0.16	0.08	0.15	0.10
	> 40	0.16	0.33	0.08	0.05	0.20	0.17
	Total	0.26	0.39	0.10	0.05	0.11	0.09

See Table 1 for infection levels description

observed in the absence of harvesting and in old partial cuts, but such relationship was not noticeable in more recent partial cuts (Fig. 4).

Discussion

American beech regeneration

The study evaluated the condition of understory AB in a region where some stands had been affected by early stages of BBD, while others were not. It thus provided us an opportunity to better understand early effects of BBD on understory regeneration dynamics.

Most striking, total AB seedlings and sprouts was less abundant in affected than in unaffected stands. AB seedlings were even three times less abundant in BBD-affected stands, while sprout density was similar between affected and unaffected stands. Moreover, no relationship was observed between BBD-induced mortality and AB sprouting.

While we found no studies directly demonstrating that BBD, in itself, triggered AB root sucker establishment, many reports have suggested that the presence of BBD would favour sprout production (Houston 1975; Hane 2003; Wagner et al. 2010; Garnas et al. 2011). Our results do not support the hypothesis that early stages of BBD necessarily triggers AB root sucker establishment in SM-AB-dominated stands. They also question the perception that dense AB thickets necessarily originate from root suckers. Jones and Raynal (1987) experimentally demonstrated that AB apical dominance is quite weak, suggesting that the loss of vigour due to BBD in a declining parent tree would have no obvious effect on sprouting. It could be argued that AB sprouting might possibly be triggered by eventual death of a tree, but this was not observed by Houston (2001).

Table 4 Model comparisons for American beech regeneration components

Component	Model	Model no.	k	AICc	Weight (ω, %)
Pooled AB seedling and sprout density	<u>BBD</u>*TBA_{AB}	1	7	1023	87.0
	<u>BBD</u>*pH + TBA_{AB}	2	8	1026	12.0
AB seedling density	<u>BBD</u>	3	4	185.5	3.3
	<u>BBD</u>*TBA_{AB}	4	7	181.9	20.2
	<u>BBD</u>*L	5	7	188	1.0
	<u>BBD</u>*TBA_{AB}*pH	6	13	187.9	1.0
	<u>BBD</u> + TBA_{AB} + L+S + pH	7	8	179.3	72.9
AB sprout density	TBA_{AB}	8	3	504.4	69.8
	<u>BBD</u>*TBA_{AB}	9	7	509.5	5.4
	TBA_{AB}*S	10	5	507.7	13.3
	TBA_{AB}*pH	11	5	508.7	8.4
	<u>BBD</u> + TBA_{AB} + S + pH	12	8	511.1	2.5
AB sapling basal area	S*T*BBD	13	25	766.3	100%

An underlined predictor variable indicate its negative effect on the response variable

AB American beech, *BBD* beech bark disease, TBA_{AB} AB tree basal area, *L* light, *S slope*, *T* harvesting treatment history, *k* number of parameters estimated in the model, *AICc* Akaike information criterion, corrected for small sample sizes. Only models with ω > 1% are shown

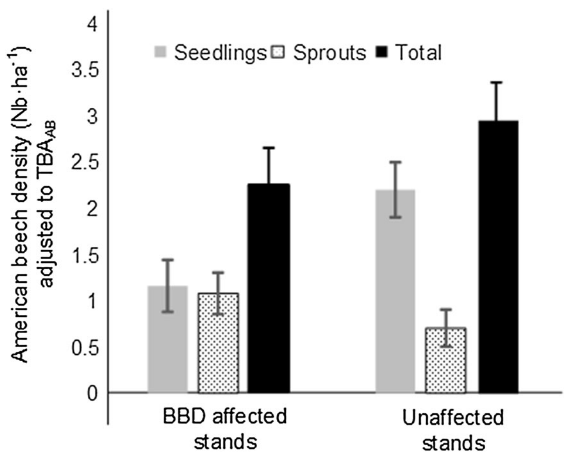

Fig. 2 Mean densities of American beech pooled seedlings and sprouts, seedlings and sprouts in BBD-affected and unaffected stands. BBD has respective cumulative probabilities (ω_c) of 99, 98.4 and 7.9% to be part of the best model to explain pooled seedlings and sprouts, seedlings and sprouts densities (see Table 4 for details). Error bars are standard errors. TBA_{AB}: American beech tree basal area (m^2 ha^{-1})

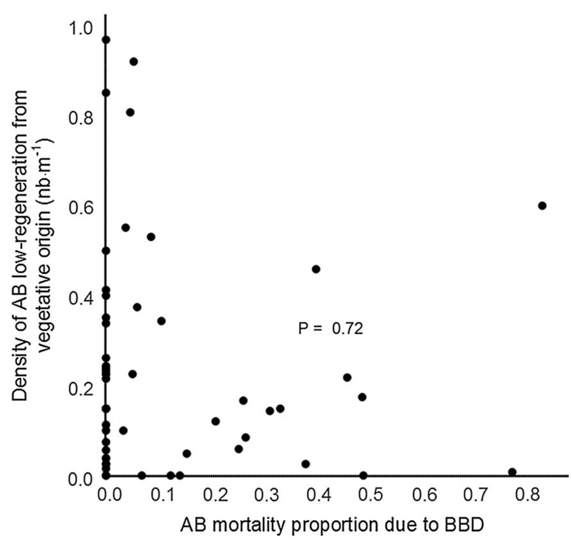

Fig. 3 Relationship between AB sprout density and AB mortality due to BBD. Each point represents a stand

Our results with regard to AB seedlings being less abundant in BBD-affected stands, agree with those of Rosemier and Storer (2010), who showed a decline in seed production for trees infected by BBD. Our findings appear inconsistent with McNulty and Masters (2005). They observed an increase in beechnut production many decades after the arrival of BBD, and that suggests a potential for increased seedlings establishment. However, this beechnut production may have originated from saplings (instead of mature trees), since AB saplings (in very high densities in

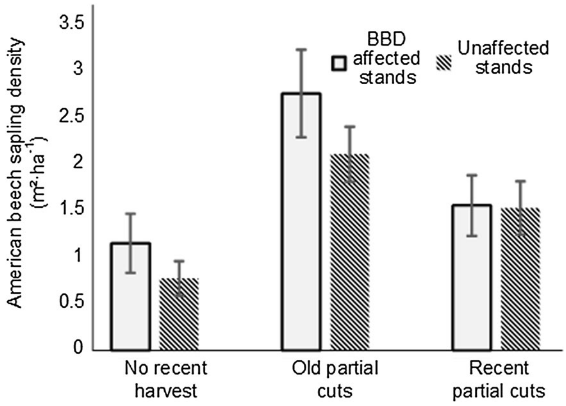

Fig. 4 Mean American beech sapling basal area in BBD-affected and unaffected stands, as a function of silvicultural treatment history. BBD and harvesting history both have a cumulative probabilities (ω_c) of 100% to be part of the best model to explain sapling density (see Table 4 for details). Error bars are standard errors

their study) as small as 3 cm DBH can produce seeds (McNulty and Masters 2005).

AB abundance at the sapling stage appears more strongly related to stand harvesting history (Fig. 4, but see also Houston 2001; Nolet et al. 2008; Bose et al. 2017) than to BBD. For that reason, we do not consider BBD as a the most influential factor for AB regeneration dominance in SM-AB-dominated stands.

While we believe that BBD (at least in its early stage) does not necessarily trigger an increase in AB sapling abundance in SM-AB stands, we do not question the idea that BBD may repeatedly favour AB regeneration over and over again through time (e.g., Houston 1975; Giencke et al. 2014; Cale et al. 2017). On one hand, we accept the evidence that AB advance regeneration is often abundant before BBD becomes established, BBD likely accelerates AB regeneration development (growth and survival) as a result of increased in understory light availability (due to upper crown reduction and mortality). That old partial cuts having greater AB sapling basal area, probably through an increase in light availability, is in accordance with the idea. On the other hand, if advance AB regeneration density is low prior to BBD infestation, our data do not support the notion that AB root suckering will neccesarily increase after the disease complex becomes established in a stand.

Management implications

The distinction between BBD acting as a trigger or merely a factor favouring AB understory development is not a trivial matter with respect to forest management. Some recommendations suggest that in face of BBD progressing northward in Québec, an appropriate management strategy would include harvesting mature trees before they become infected by BBD and produce sprouts. Our results suggest that such a strategy may not be necessary, given that BBD role in AB sprout production is unclear for the Outaouais region. Rather, we suggest removing understory beech (Nyland 2004; Ostrofsky 2005; Nolet et al. 2015), when abundant. Mature trees could be harvested, but could be left for ecological reasons or to serve other ownership objectives.

Conclusion

Our findings do not implicate BBD, in itself, as the trigger for root suckering in AB-SM stands of Quebec. They do show that a reduction in density of the upper canopy following onset and progression of the disease does appear to accelerate understory sapling development and promote AB dominance in a stand. Consistent with that, unaffected stands that had past partial cuts also had a greater degree of AB understory development than stands not yet harvested. Data also show that stands not affected by BBD have more AB seedlings than stands with the disease complex, and that seedlings outnumber root suckers in unaffected stands. Overall, our results question the generalized perception that dense AB thickets necessarily originate from root suckers in BBD affected stands. More research is required to quantify how and when production of both seedlings and sprouts is affected by BBD.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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